Random Forest Classification of Physiological Signals for Mental Health Screening in Learning Institutions

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1 Introduction

Mental Health disorders are a growing problem and major health factor all over the world. 700000 suicide cases happen globally every year and 77% of these happen in low and middle income countries like Uganda [1]. Factors such as stress, anxiety and depression play pivotal roles as they re-construct the brains reward system and make one lose interest in themselves.

Within Uganda mental health is a huge concern among the young people. Estimates show that 23.6% of Ugandan children live with depression [2]. Most of these are school going and spend most of their time at schools yet no mental health assessment is included. Physical condition like height, weight BMI are measured but rarely is mental health assessed. As a result stress, anxiety and depression remain undetected leading to fatal outcomes.

Despite this, mental health screening in schools is low. Councilors exist but underutilized. Traditional methods like questioners have been used though are prone to bias as students rarely disclose the truth out of fear. Advanced methods such as EEG are reliable, but expensive and need professionals to help out analysis. Other studies, brought up the idea of using wearable sensors but were prone to noise by movement which could disrupt data collection [3].

Recent studies have showed pupil dilation and constrictions as possible biomarkers for brain activity. The pupil dynamics are controlled by the autonomic nervous system that has two branches namely sympathetic and parasympathetic systems. The sympathetic is triggered during stress and pupils dilate while the parasympathetic induces constriction as body relaxes [4].The hypothalamus, connects the endocrine system and nervous system ,hence changes in emotion lead to cortisol being released leading to increase in blood pressure and heart rate [5]. Together these mechanisms are a gateway to the mind, and when combined with heart rate and blood pressure, they provide a multi modal picture of mental health.

This study aims at addressing the gap of poor mental health screening in schools by introducing a Random forest based screening system that will combine, pupil di-

amteres, blood pressure and heart rate. By integrating multiple signals, the model will provide more reliable assessment rather than relying on one metric.

References

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